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STATE-OF-THE-ART REVIEW

Potential roles of super enhancers in inflammatory gene transcription

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Acute and chronic inflammation is a basic pathological event that contributes to atherosclerosis, cancer, infectious diseases, and immune disorders. Inflammation is an adaptive process to both external and internal stimuli experienced by the human body. Although the mechanism of gene transcription is highly complicated and orchestrated in a timely and spatial manner, recent developments in next-generation sequencing, genome-editing, cryo-electron microscopy, and single cell-based technologies could provide us with insights into the roles of super enhancers (SEs). Initially, SEs were implicated in determining cell fate; subsequent studies have clarified that SEs are associated with various pathological conditions, including cancer and inflammatory diseases. Recent technological advances have unveiled the molecular mechanisms of SEs, which involve epigenetic histone modifications, chromatin three-dimensional structures, and phase-separated condensates. In this review, we discuss the relationship between

Abbreviations
3C, chromosome conformation capture; ATAC-seq, assay for transposase-accessible chromatin using sequencing; BACH2, BTB domain and CNC homolog 2; BET, bromodomain and extra-terminal; BRD2/3/4, bromodomain containing 2/3/4/; BRDT, bromodomain testis associated; BRG1/SMARCA4, SWI/SNF related, matrix associated, actin dependent regulator of chromatin, subfamily a, member 4; C/EBPα, CCAAT enhancer binding protein alpha; CBP, CREB binding protein; CCL, C-C motif chemokine ligand; CDK, cyclin-dependent kinase; ChIA-PET, chromatin interaction analysis by paired-end tag sequencing; ChIP-seq, chromatin immunoprecipitation followed by sequencing; CKIα, casein kinase 1A1; CRISPR, clustered regularly interspaced short palindromic repeats; CTCF, CCCTC-binding factor; CTD, C-terminal domain; CXCL, C-X-C motif chemokine ligand; EBF1, EBF transcription factor 1; EDN1, endothelin 1; ERG, ETS-related gene; eRNA, enhancer RNA; ES, embryonic stem; Esrrb, estrogen related receptor beta; GATA2, GATA binding protein 2; GR, nuclear receptor subfamily 3 group C member 1; GTF, general transcription factor; H3K27ac, acetylation of lysine 27 in histone H3; H3K27me3, trimethylation of lysine 27 in histone H3; H3K4me1, monomethylation of lysine 4 in histone H3; H3K4me2, dimethylation of lysine 4 in histone H3; H3K4me3, trimethylation of lysine 4 in histone H3; H3K9me2, dimethylation of lysine 9 in histone H3; HICE, highly interconnected enhancers; HNF4A, hepatocyte nuclear factor 4 alpha; IDRs, intrinsically disordered regions; IFN, interferon; IgH, immunoglobulin heavy chain; Ikaros, IKAROS family zinc finger 1; IL, interleukin; IRF4, interferon regulatory factor 4; JAK, Janus kinase; KDM7A, lysine demethylase 7A; Klf4, Kruppel like factor 4; LncRNA, long noncoding RNA; LPS, lipopolysaccharide; MAF, MAF bZIP transcription factor; Med12, mediator complex subunit 12; Myc, MYC proto-oncogene, bHLH transcription factor; Nanog, Nanog homeobox; NF-κB, nuclear factor kappa B; Oct4, POU class 5 homeobox 1; p300, E1A binding protein p300; Pax5, paired box 5; pol II, RNA polymerase II; PROTAC, proteolysis-targeting chimera; P-TEFb, positive transcription elongation factor b; PU.1, Spi-1 proto-oncogene; ROSE, ranking of super enhancer; RUNX1, RUNX family transcription factor 1; SE, super enhancer; SNP, single nucleotide polymorphism; Sox2, SRY-box transcription factor 2; STAT, signal transducer and activator of transcription; T/ brachyury/TBXT, T-box transcription factor T; TADs, topologically associating domains; T-bet, T-box transcription factor 21; TFIIH, transcription initiation factor IIH; Th, T helper; TLR, toll like receptor; TNF-α, tumor necrosis factor alpha; UMLILO, upstream master lncRNA of the inflammatory chemokine locus; UTX, lysine demethylase 6A; VCAM1, vascular cell adhesion molecule 1.

inflammation and SEs and the therapeutic potential of SEs for inflammatory diseases.

Introduction

Enhancers are classically defined as transcriptional regulatory elements located away from the promoter or within the gene body which function independently of orientation [1]. In general, enhancer regions can directly activate transcription via acting on the associated promoters and are crucial for cell-type-specific gene transcription [2,3]. Mechanistically, pause-release (i.e., productive elongation) of RNA polymerase II (pol II) has been considered as a major point of enhancer-mediated transcriptional activation, albeit a series of very recent studies has implicated the functional importance of enhancer in recruitment and transcriptional initiation of pol II [4–6]. Current evidence has shown that enhancer regions are associated with DNase I hypersensitivity (i.e., open chromatin), binding of transcription factors, recruitment of co-regulators (e.g., Mediator, acetyltransferase CBP/p300), epigenetic modifications [e.g., H3K4me1/2 (primed enhancers), H3K27ac (active enhancer)], and chromatin architecture, which contribute to the transcription of enhancer RNAs (eRNAs) [7,8] (Fig. 1). With the expanding concept of enhancers, 'super enhancers (SEs)' including clustered and stretched enhancers have also been proposed. SEs are large clusters of enhancers harboring unusually high level of transcriptional regulators and believed to have a stronger correlation with cell specificity than typical enhancers [9–11] (Fig. 1). Although most studies on SEs to date have been performed in undifferentiated cells and cancer cells, there are strong indications that these are also involved in other biological processes, such as inflammation, immunity, and neuronal degeneration, given their roles as determinants of cellular properties [12]. However, the relationship between SEs and various cellular and molecular functions remains largely unclear. On the other hand, several groups have recently reported even in terminally differentiated cells that SEs and their related chromatin architecture (3D genome organization) can be immediately altered in response to inflammatory stimuli, resulting in aberrant expression of inflammatory genes [13–15]. Importantly, SEs and SE-related genes are highly vulnerable to inhibition of enhancer factors, suggesting the therapeutic potential of SE inhibition in inflammatory diseases [16]. Here in this review, we describe the current insights on gene regulatory mechanisms involving SEs focusing on inflammation. We further discuss recent advances and challenges of SE inhibition for the treatment of inflammatory diseases.

Definition and properties of SEs

Super enhancers differ from typical enhancers in length and transcription factor density. The concept of SEs was first described in mouse embryonic stem (ES) cells and several tumor cell lines [9,17]. SEs are defined as stretched genomic regions that are highly bound with five master transcription factors (Oct4, Sox2, Nanog, Esrrb, and Klf4) and Mediator. Several studies have expanded the concept of SEs, indicating that these not only contain several transcription factors but also characteristic histone modifications, such as H3K27ac and H3K4me1, and p300 in various cell types and tissues [10,12,17,18]. SEs in mouse ES cells were first defined using the following method [9]: (a) regions containing all the three master transcription factors Nanog, Oct4, and Sox2 identified via ChIP-seq were considered as enhancers; (b) enhancers within 12.5 kb of each other were combined and defined as a single entity; (c) the combined enhancer entities and the remaining individual enhancers were then ranked based on the total background-normalized level of Med1 signal within the genomic region. Next, the main identification procedure involved the following steps: (a) obtaining the ChIP-seq data for H3K27ac, (b) combining enhancers within 12.5 kb of each other, based on calculated ranking of SEs (ROSE); (c) plotting the combined enhancers according to ranks and the remaining individual enhancers according to the total enrichment levels of H3K27ac within the genomic region [9,17]. In other words, SE is determined by the aforementioned process and not by functional definition.

Super enhancers, as defined above, tend to occupy genomic regions many times longer than those of typical enhancers; for instance, in mouse ES cells, average length of SEs is 8667 bp whereas that of a typical enhancer is 703 bp [9,10,17]. Moreover, SE regions generally produce higher levels of eRNAs than typical enhancers.

In addition, SEs are thought to have an important role in determining cell specificity. Previous studies

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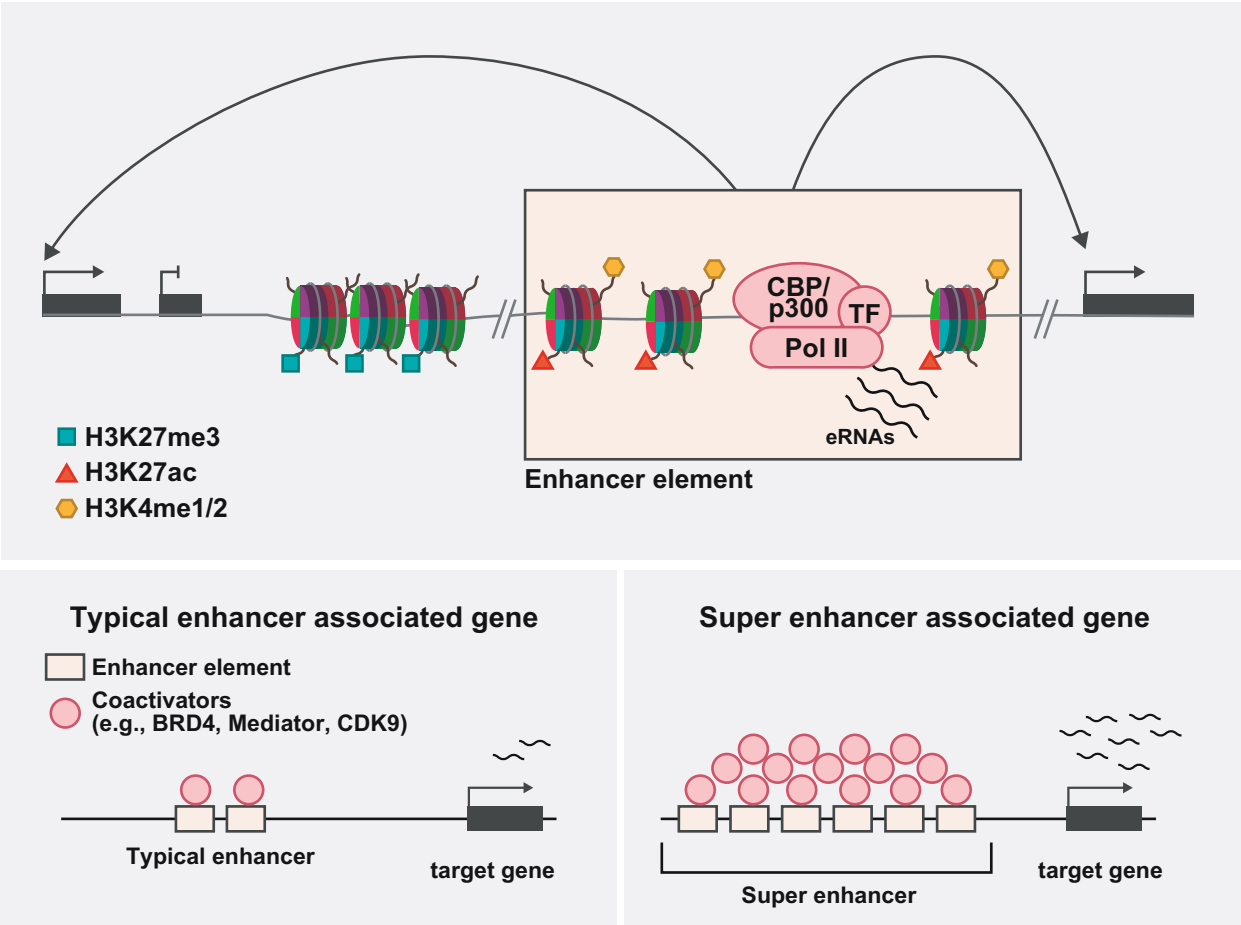

Fig. 1. Current understanding of features of enhancer and super enhancer. Enhancers are defined as cis-acting DNA sequences that can increase the transcription of genes. Enhancers generally function independently of orientation and at various distances from their target promoter. Enhancers are characterized by (a) DNAse I hypersensitivity (i.e., open chromatin), (b) binding of transcription factors (TFs), (c) recruitment of coactivators (e.g., CBP/p300 and Mediator), (d) transcription of enhancer RNA (eRNA), and (e) deposition of specific histone marks [e.g., H3K4me1/2 (primed enhancer) and H3K27ac (active enhancer)] (upper panel). SEs are large clusters of enhancers with unusually high levels of binding of coactivators [e.g., bromodomain containing 4 (BRD4), Mediator, and cyclin-dependent kinase 9 (CDK9)], which are strongly correlated with cell-type-specific transcription compared with typical enhancers (lower panel).

have shown that transcription factors are evidently responsible for regulating cell-type-specific gene expression [19–21], and master transcription factors accumulate in SE regions. For example, some groups have investigated binding regions for five master transcription factors (i.e., Nanog, Sox2, Oct4, Klf4, and Esrrb) in mouse ES cells; of these, Nanog, Sox2, and Oct4, which are important for pluripotency, are clustered in the typical enhancer regions, whereas Klf4 and Esrrb are enriched in the SE regions, where strong binding with Mediator is observed. Knockdown of the Mediator subunit Med12 specifically affects the expression of genes adjacent to SEs but not of those adjacent to typical enhancers [9,22]. Thus, SE regions are crucial for ES cell identity. Likewise, master transcription factor binding SEs are important for cell identity in other cell types such as macrophages, epithelial cells, hair follicle stem cells, adipocytes, and helper T cells [23–26].

Super enhancers accumulate transcription factors and transcriptional complexes; however, the molecular mechanisms that regulate several cell-type-specific gene expression are largely unknown. A new phase separation model for SEs and cell-specific gene transcription was proposed in 2017, wherein proteins and nucleic acid components of enhancers and SEs exist as chain molecules, each containing intrinsically disordered regions (IDRs) with residues that may interact with other chains. The residues on the chain molecules may

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undergo reversible chemical modifications related to their ability to form noncovalent cross-links between the chains [27]; a cooperative process of phase separation involving an ensemble of multivalent interactions of IDRs with transcription factors, coactivators, Mediator, and RNA pol II can lead to dynamic condensation of these factors. They can be aggregated into specific enhancer elements [28–31]. However, the direct association between inflammatory diseases and phase-separated SEs has not been intensively studied until now. Given that SEs play crucial roles in inflammatory diseases, as discussed in the following sections, studies linking phase separation, SEs, and inflammation are expected to increase considerably in the near future.

SEs in immune cells and nonimmune cells

Since the original concept of SEs was developed in ES cells, SEs have been described in several other cell systems. SEs and their associated genes are particularly sensitive and dynamic with respect to external stimuli, including inflammatory cytokines, regardless of their characteristic features for engaging large transcriptional machinery such as transcription factors, active histone marks (e.g., H3K27ac), Mediator, RNA pol II, and acetyltransferase p300/CBP. Considering that immune cells are plastic even in their mature state, it is reasonable to explore the association of SEs with maintenance of immune cell identity and their further differentiation into specific subtypes in response to extracellular stimuli. An early study defined SEs in several immune cells using a genome-wide binding profile of lineage-determining transcription factors: PU.1 in pro-B cells, T-bet in helper T cells, and C/EBPα in macrophages [9]. In these immune cells, SEs spanned large genomic domains in a cell-type-specific manner in association with the genes that feature predominantly in the biology of these cells, supporting the idea that SEs containing cell-type-specific transcription factors play an important role in the control of immune cell identity [9]. Macrophages reside within different tissues and adapt to their surrounding environment, resulting in diverse populations that have tissue- or niche-specific functions. Despite common features shared across all subtypes (e.g., dependency on master transcription factor PU.1), each subtype of tissue macrophages possesses its own unique gene expression profile, suggesting the utilization of different enhancer repertoires depending on their surrounding environment. Gosselin *et al.* compared the gene expression and chromatin landscape among five populations of macrophages: microglia (brain macrophages), three subtypes of peritoneal macrophages, and bone marrow-derived macrophages. While ~ 600 to 750 SEs, as defined using H3K27ac ChIP-seq, are common among the five macrophage subsets, approximately 40–50% of the SEs in a particular macrophage subset are unique to that subset, associated with genes that are highly differentially expressed in that subset. Interestingly, SEs unique to a particular macrophage subset were marked by H3K4me2 in the other macrophage subsets, suggesting that environmental signals may be an important cue in the transition of primed enhancers to active SEs for developing into a distinct macrophage subset [26]. Another group defined SEs in the Th1, Th2, and Th17 subsets of CD4+ T helper cells. Consistent with a previous report [26], ChIP-seq for p300 demonstrated that approximately half of the SEs were common among these three subsets but the remaining half were unique and mostly related to cytokines, cytokine receptors, and lineage-determining factors that were characteristic to their own subset. Notably, the locus encoding *Bach2*, a key negative regulator of effector differentiation, was the strongest SE region in all Th1, Th2, and Th17 subsets, indicating a negative feedback system where genes critical for T-cell biology can be repressed hierarchically by SE-regulated BACH2 [32]. Thus, SEs can be clearly different even among closely related subpopulations of immune cells. ATAC-seq combined with binding profiles of multiple lineage-determining transcription factors revealed that during B-cell development, the ATP-dependent chromatin remodeler Brg1 is essential for maintaining chromatin accessibility of the distal SE of Myc and enabling the binding of key transcription factors such as EBF1, Ikaros, and Pax5 for activating Myc expression and promoting pro-B-cell growth [33]. Additionally, the distal enhancers (identical to SE) of Myc have been implicated in acute myelogenous leukemia, in which Brg1 plays a critical role in maintaining transcription factor occupancy and long-range chromatin looping interactions with the Myc promoter [34]. The H3K27me3 histone demethylase UTX facilitated the accessibility of SEs establishing the cell identity and thereby controlled the specific gene expression program during the development of invariant natural killer T cells in a demethylase-activity-dependent manner [35].

In terms of SE-responsiveness against inflammatory stimuli in immune cells, *de novo* chromatin accessibility and the associated rapid remodeling of SEs have been demonstrated in natural killer cells, both *in vitro* and in a mouse model of *Toxoplasma gondii* infection, in which signal-induced STAT family transcription factors initiate chromatin opening with redeployment

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of the lineage-determining transcription factor T-bet [36]. Treatment of the human macrophages with interferon (IFN)-γ suppressed basal expression of genes corresponding to an 'M2'-like homeostatic and reparative phenotype, which was mediated through suppressing the function of enhancers enriched for binding by transcription factor MAF. Mechanistically, IFN-γ disassembled a subset of enhancers by inducing coordinate suppression of binding by MAF, lineage-determining transcription factors, and chromatin accessibility. Importantly, the *MAF* locus itself was associated with a SE harboring multiple H3K27ac and ATAC-seq peaks, which was effectively abolished by IFN-γ treatment [37]. Furthermore, SEs and SE-associated eRNA transcription dynamics are associated with both the activation and repression of gene expression during the activation of TLR4 signaling by lipopolysaccharide (LPS) treatment in macrophages. Genes associated with SEs gained after LPS treatment were primarily involved in either innate immunity or inflammation, whereas those associated with lost SEs mostly exhibited general functions in cellular metabolism and nuclear organization [38]. A series of studies have shown that eRNA expression and enhancer activity precede transcriptional induction of inflammatory genes in response to LPS [39–41]. More recently, tamoxifen-induced ablation of lineage-determining transcription factor GATA2 has revealed that GATA2 regulates mast cell identity and responsiveness to antigenic stimulation by promoting chromatin remodeling at SEs [42]. Interestingly, in mouse primary B lymphocytes upon B-cell receptor activation, a mathematical modeling revealed that higher fold change of mRNA induction is associated with DNA length of SE while threshold response (i.e., switch-like) response is predefined by the coexistence of PU.1 and nuclear factor (NF)-κB in the SE [43]. Collectively, these studies suggest that SEs are critical genomic modules for not only maintaining cell identity but also responding to external environmental signals in immune cells.

Super enhancer characterizations in inflammation have extended beyond immune cells. Recent studies have implied that SEs in nonimmune cells function in response to inflammatory stimuli, likely involving cell state transitions. A study of human endothelial cells demonstrated that stimulation with tumor necrosis factor (TNF)-α resulted in the formation of signal-induced NF-κB-bound SEs co-occupied by H3K27ac and its reader protein, bromodomain containing 4 (BRD4), within 1 h following treatment [44]. Disruption in the formation of SEs by JQ1, a BET bromodomain inhibitor, strongly inhibited TNF-α-induced inflammatory gene transcription, suggesting potential roles of SEs as driving forces underlying the immediate activation of inflammatory signaling. On the contrary, genes rapidly repressed by TNF-α signaling are associated with SEs where BRD4 binding is abrogated by TNF-α. Importantly, these repressed genes mostly belong to vascular development or maintenance gene categories, indicating that SEs play important roles in both maintaining vascular homeostasis as well as shifting from 'resting' to 'activating' states in protective immune responses. The same phenomena (i.e., acute gain and loss of SEs) have also been reported in TNF-α-stimulated human adipocytes in which cell identity genes are rapidly repressed through transcriptional cofactors- and eRNA-redistribution from basal SEs to TNF-α-induced NF-κB-bound SEs [45]. A recent study revealed that, in human endothelial cells, lysine demethylase 7A (KDM7A) and UTX are rapidly mobilized to TNF-α-induced SEs and that demethylation of the repressive histone marks H3K9me2 and H3K27me3 by KDM7A and UTX, respectively, was required for the transcriptional activation of NF-κB-dependent inflammatory genes. Moreover, genomic binding of KDM7A and UTX was rapidly de-localized from resting SEs (related to vascular homeostasis genes) in response to TNF-α stimuli despite the persistent hypomethylation of H3K9me2 and H3K27me3, respectively. Simultaneous pharmacological inhibition of KDM7A and UTX reduced leukocyte adhesive events in mice [13]. Moreover, during kidney repair after ischemia–reperfusion injury (proximal tubular epithelial cells are mostly damaged), key transcription factors including HNF4A, GR, STAT3, and STAT5 form SE structures for tissue repair, revealing enhancer dynamics and transcriptional changes during kidney repair processes. JQ1-treatment before ischemia–reperfusion injury leads to impaired recovery and increased mortality [46].

Collectively, these findings indicate that SEs are much more dynamic and sensitive to changes in the external milieu, even in completely differentiated cells, suggesting their potential as therapeutic targets for various inflammatory diseases, including cardiovascular diseases and cancer (Fig. 2). Supporting this idea, a recent study has shown that advanced renal cell carcinoma reshapes the immune landscape to secrete cytokines and chemokines (cause of systemic deleterious inflammation) via epigenetic remodeling through mechanisms of DNA demethylation and SE-formation; in addition, bromodomain inhibition by JQ1 suppressed chemokine secretion and decreased neutrophil-dependent lung metastasis [47].

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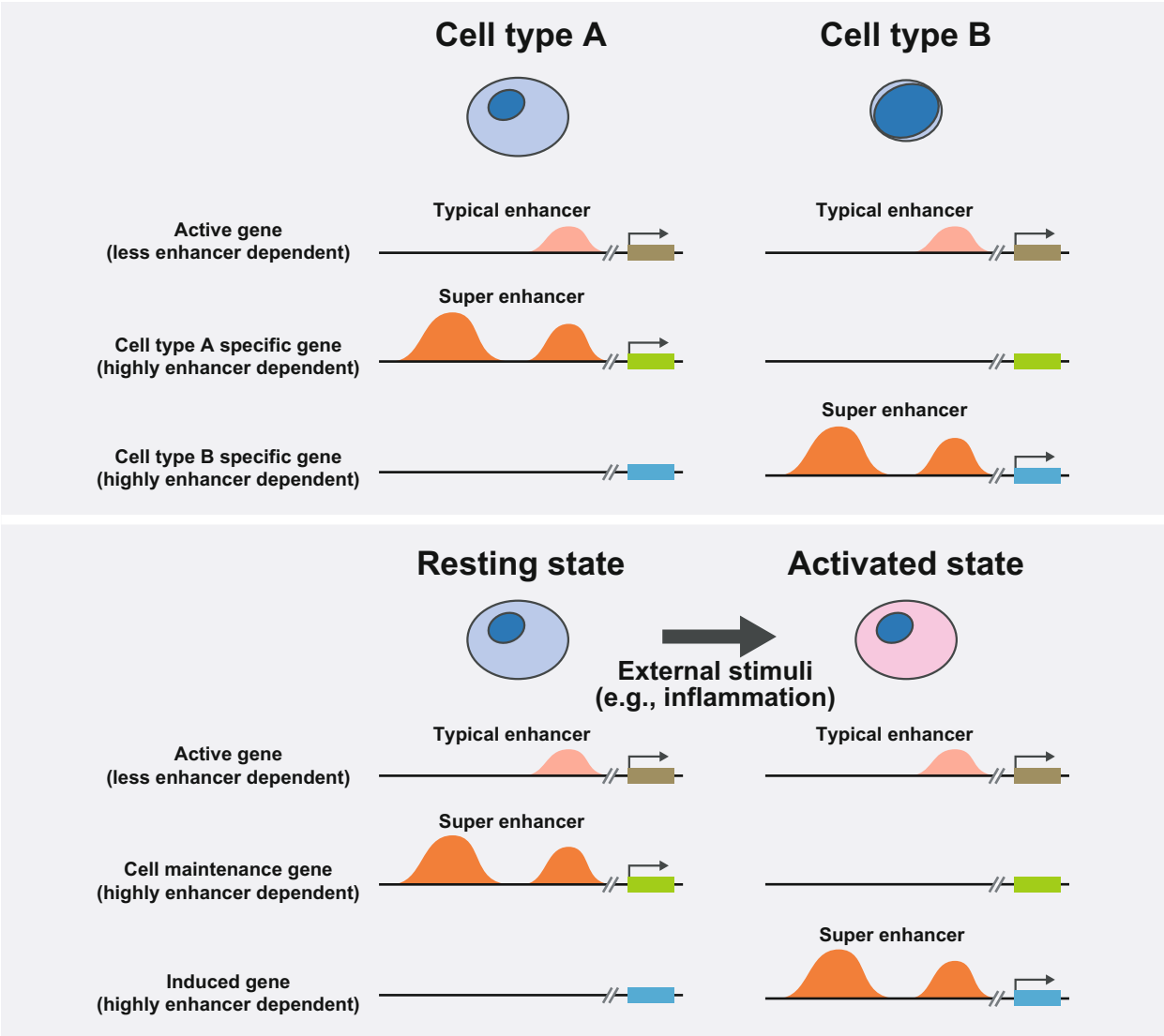

Fig. 2. Super enhancers function in both maintaining the cell-type-specific gene expression and signal-dependent transcriptional regulation. SEs span larger domains occupied by multiple transcriptional activating proteins such as TFs and Mediator than typical enhancers (hypothetical gene tracks are shown). SEs drive cell-type-specific gene expression (upper panel). Additionally, SEs are involved in signal-dependent transcriptional regulation including inflammatory response (lower panel).

SEs and 3D genome in inflammatory gene transcription and diseases

Advances in chromosome conformation capture (3C)-derived sequencing technologies, such as Hi-C and chromatin interaction analysis by paired-end tag sequencing (ChIA-PET), have enabled systematic analysis of genome-wide interactions of chromatin and revealed important roles of the 3D organization of the genome in transcriptional control [48]. Hi-C has revealed that the genome is compartmentalized into large self-interacting genomic regions, termed as topologically associating domains (TADs), which could be associated with either an active or inactive state of chromatin. Although TADs are already formed in ES cells and remain relatively stable during development/differentiation as well as among cell types [49,50], recent studies have suggested that genome organization can greatly vary even between cells of the same individual [51] and that chromatin loops within large megabase-sized TADs (i.e., subTADs) are clearly different among cell types and can vary in response to external stimuli, alongside playing instructive roles in gene regulation [52,53].

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Remarkably, genomic architecture mapping, a new ligation-free method for capturing 3D proximities based on DNA sequencing data from a large collection of thin nuclear sections, demonstrated that SEs are highly enriched among the genomic regions with high frequencies of chromatin interactions [54]. Notably, when SEs are ranked according to the frequency of chromatin interactions based on Hi-C data, hierarchical SEs (highly interacting SEs) are more likely to be associated with the gene ontology terms for immune and defense responses than nonhierarchical SEs. The most highly interacting regions within hierarchical SEs can be associated with cohesion and CTCF binding sites as well as disease-associating variants; CRISPR-mediated ablation of these regions caused marked defects in gene activation and local chromatin conformation, suggesting essential roles of 3D chromatin organization in orchestrating inflammatory activities of SE [55]. Previous studies have shown that most TNF-α-induced genes are prelooped between genomic enhancer and enhancer/promoter regions in human endothelial cells and can be relatively constant during TNF-α signaling [56,57]. In contrast, using Hi-C in combination with ChIA-PET, our group revealed the emergence of new subTAD scale interactions among SEs associated with typical TNF-α-responsive inflammatory gene loci in human endothelial cells in response to TNF-α stimuli. Importantly, the ChIA-PET interactions increased both at the KDM7A- and UTX-binding sites in TNF-α-treated endothelial cells, in conjunction with decreased levels of the repressive histone marks H3K9me2 and H3K27me2 [13]. Another group analyzed multiple datasets of Hi-C, ChIA-PET, cap analysis gene expression, and ChIP-seq data, and identified a novel SE-associated long noncoding RNA (lncRNA), an upstream master lncRNA of the inflammatory chemokine locus UMLILO. UMLILO is located on and transcribed from the SE region within the chemokine genes in clustered TAD, including *IL-8*, *CXCL1*, *CXCL2*, and *CXCL3* loci. Notably, UMLILO is highly expressed in innate immune cells and functions in cis to direct WD repeat-containing protein 5 and mixed lineage leukemia protein 1 complex toward the chemokine gene promoters, facilitating active histone mark H3K4me3-mediated epigenetic priming of inflammatory transcriptional responses. Of note, UMLILO depletion abrogated TNF-α-induced chemokine expression, which was associated with a modest but not significant decrease in chromosomal interactions among the chemokine genes in clustered TADs. Insertion of UMLILO into the genomic region containing mouse chemokine genes in clustered TAD (UMLILO is absent from the mouse genome) resulted in H3K4me3-mediated priming of chemokine gene transcription [14]. Furthermore, a recent study using enhancer-capture Hi-C has revealed that approximately half of the known SEs in adipocytes overlap with highly interconnected enhancers (HICE), and genes interacting with both SE and HICE are much more dynamic than those interacting with either SE or HICE [58].

3C-based sequencing has also been used as a powerful tool for physically linking noncoding variants of genes causing inflammatory diseases to their molecular target [59]. H3K27ac-HiChIP recently revealed that noncoding variants related to inflammatory dermatological diseases were assigned a 3.6-fold increased number of target genes, but those related to noninflammatory dermatological diseases were assigned a 0.6-fold decreased number of target genes when enhancer–promoter interaction-based assignment was applied instead of nearest gene assignment [60]. Consistently, in human islets, hub enhancers defined by both 3D interactions and high occupancies of H3K27ac and Mediator were more effective in predicting tissue-specific risk scores for type 2 diabetes than SEs (without considering 3D interaction), despite the fact that hub enhancers overlap with 87% of SEs [61], suggesting the possible importance of enhancer/SE-influenced genome organization in inflammatory diseases. For instance, a noncoding variant associated with abdominal aortic aneurysm, rs2836411, resides within the SEs of the ETS-related gene (*ERG*) in vascular endothelial and hematopoietic cells and is associated with reduced ERG expression. Allele-specific 4C-seq demonstrated that the risk allele has fewer interactions within the SE compared to the protective allele [62]. Genetic fine mapping identified a common SNP, rs9349379, located on the third intron of the phosphatase and actin regulatory protein 1 (*PHACTR1*) gene, as a putative causal variant of five vascular diseases including coronary artery disease and hypertension. In CRISPR-edited stem cell-derived human endothelial cells, the introduction of rs9349379 increased the expression of endothelin 1 (*EDN1*), a gene located 600 kb upstream of *PHACTR1*, which could explain the risk pattern of the five associated diseases. Interestingly, 4C-seq revealed a very low contact probability between rs9349379 and the *EDN1* promoter, but a relatively high contact probability between rs9349379 and the *EDN1* downstream SEs marked with high levels of H3K27ac, suggesting a potential role of rs9349379 in SE function [63]. Furthermore, cardiovascular disease-related variants are highly enriched in vascular endothelial growth factor-regulated genomic compartments. The variant

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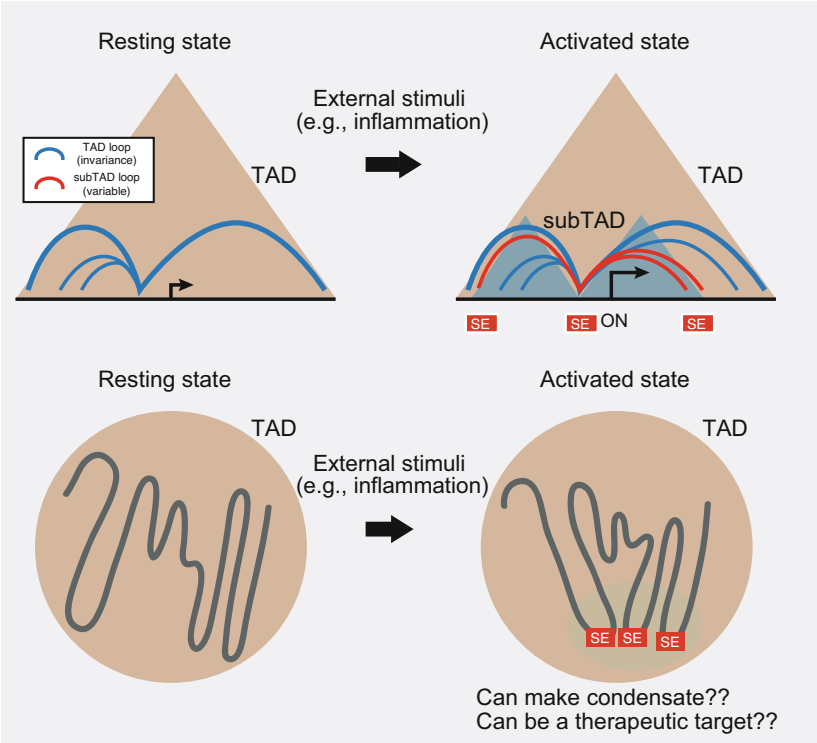

Fig. 3. Chromosomal conformation changes in response to external stimuli. Topologically associating domains (TAD) are relatively stable; however, chromatin loops within TAD (i.e., subTAD/subTAD loop) can vary in response to external stimuli including inflammation (upper panel). Recent studies have suggested that SEs participate in forming phase-separated condensates; however, it remains unknown whether this type of model could be applied in signal-dependent transcription mechanisms and phase separation could be a therapeutic target for inflammatory diseases (lower panel).

rs1371799 is located in the SE-loop of the *IL-8* locus, which exhibits rapid chromosomal conformation change in response to TNF-α stimulus [13,64].

Collectively, these findings suggest that proper 3D genome chromatin configuration may be crucial for SE-mediated transcription regulatory functions, and disruption of this genomic organization might be a potential cause of common inflammatory diseases (Fig. 3).

Clinical application of enhancer/SE-inhibitors in inflammatory diseases

Although it could be practically impossible to specifically target SEs using inhibitors, SE-related genes are exceptionally vulnerable to the inhibition of enhancer factors; thus, it might be possible to preferentially target SE-related genes using inhibitors. The underlying mechanism of this vulnerability is not fully understood; multiple models have been proposed to explain this phenomenon [16]. Indeed, the Janus kinase (JAK) inhibitor tofacitinib, which is used to treat rheumatoid arthritis, disproportionately altered the expression of rheumatoid arthritis risk genes associated with the SE structure in T cells [32]. On the basis of these concepts, extensive efforts have been made to develop new therapeutic strategies targeting enhancer factors for the treatment of various human diseases, including cancer and inflammation. At present, inhibitors of bromodomain and extra-terminal (BET) and cyclin-dependent kinase (CDK) have emerged as promising targeting approaches, especially for cancer therapy, possibly through modulating the transcription of SE-related genes.

In 2010, two independent innovative studies showed that the highly potent inhibitors JQ1 and I-BET762 (GSK525726A) selectively target the BET family proteins BRD2, BRD3, BRD4, and BRDT by interfering with the abilities of these proteins to recognize acetylated histones, thus resulting in robust tumor-suppressive effects on cancer cells with BRD4-fusion oncoproteins, strong suppression of LPS-induced expression of inflammatory genes in macrophages, and remarkable improvement in the survival rates of *in vivo* sepsis models [65,66]. Although enhancers generally function as transcriptional activators by promoting transcriptional elongation (i.e., RNA pol II pause-release [67]), BET inhibitors have been shown to affect the transcription of a relatively small subset of genes. Treatment of multiple myeloma MM1.S cells (overexpressing MYC as a consequence of *IgH-MYC* rearrangement) with JQ1 led to preferential loss of BRD4 at SEs and consequent transcriptional elongation defects in SE-related genes. Treatment of MM1.S cells

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with even lower concentrations of JQ1 dramatically reduced BRD4 occupancy at the exemplified SE region of the *IgH-MYC* locus, whereas loss of BRD4 occupancy was modest at the typical enhancer region of *SMARCA4*. Higher concentration of JQ1 abrogated virtually all BRD4 occupancy at the SE region of *IgH-MYC*, while BRD4 occupancy was maintained at the typical enhancer region of *SMARCA4*. SE-associated oncogenes such as MYC and IRF4 were more rapidly and strongly decreased than highly expressed genes with typical enhancer after the treatment with JQ1 [17]. Notably, exceptional sensitivity of SE-associated oncogenes to a BET inhibitor was observed in various cancers [68,69]. Consistent with these reports, treatment of TNF-α-stimulated endothelial cells with JQ1 resulted in preferential loss of BRD4 at SEs compared with that at typical enhancers. JQ1-treatment significantly decreased SE-associated proinflammatory genes including VCAM1 and CCL2 in comparison with the genes associated with typical enhancers. Notably, treatment of atherogenic mice with JQ1 significantly reduced aortic plaque area and infiltration of immune cells such as T lymphocytes and macrophages, accompanied by reduced expression of inflammatory adhesion molecules, which suggested the therapeutic potential of BET bromodomain inhibition for systemic proinflammatory activation *in vivo* [44]. Furthermore, disease-specific, inflammation-associated SEs were identified in synovial fluid-derived CD4 T cells from a juvenile idiopathic arthritis patient; BET inhibition of the patient-derived T cells with JQ1 preferentially reduced disease-associated inflammatory gene expression [70]. A recent study also revealed that SEs at the *Il9* locus are critically required for OX40-mediated robust induction of Th9 cells and disruption of SEs by JQ1 suppressed allergic airway inflammation in mice [71].

Currently, there are at least 10 ongoing clinical trials, excluding cancer trials [72] evaluating the efficacy or safety of BET inhibitors, for several human diseases including type 2 diabetes and coronary artery disease. For example, a phase III clinical trial (BETonMACE) to test the protective effects of apabetalone (RVX000222), a novel orally active BET inhibitor, in high-risk type 2 diabetes patients with coronary artery diseases is in progress, whereas an interim report showed that apabetalone did not substantially reduce the risk of major adverse cardiovascular events [73]. Moreover, proteolysis-targeting chimera (PROTAC) of BET proteins, which can selectively degrade BET inhibitor-binding target molecules, has also emerged as a new type of drug [72]. A Phase 1/2 clinical trial of ARV-110, an orally bioavailable PROTAC targeting androgen receptor in prostate cancer patients, has already been started (https://clinicaltrials.gov/ct2/show/NCT03888612).

Beyond its roles in cell cycle regulation, transcriptional CDKs have been recognized as compelling targets for cancer therapy. CDK7 is a component of transcription initiation factor IIH (TFIIH) in the general transcription factor (GTF) complex. CDK7 phosphorylates the C-terminal domain (CTD) of the largest subunit of Pol II at Ser5 and Ser7, which triggers Po lI release from promoters, thereby allowing transition from initiation to productive elongation of Pol II. CDK7 additionally phosphorylates and activates CDK9 (a catalytic subunit of positive transcription elongation factor b (P-TEFb) complex), which phosphorylates Ser2 in the CTD of Pol II, a modification that controls elongation and termination [74]. P-TEFb is present in the BRD4-containing protein complex where BRD4 promotes recruitment of P-TEFb to the promoters of the genes through binding to the acetylated chromatin or Mediator complex [75]. In T-cell acute lymphoblastic leukemia cells, SE regions of *RUNX1* display exceptionally high occupancies of CDK7, and small-molecule inhibition of CDK7 by THZ1 (also potentially targeting CDK12 and CDK13) leads to preferential loss of transcripts for oncogenic transcription factors including *RUNX1* [76]. The preferential downregulation of SE-associated genes by THZ1 was confirmed in various cancers [77–79]. More recently, genome-scale CRIPSR screening identified transcriptional CDKs as potential therapeutic targets for chordoma, and inhibition of CDK7 and CDK9 by THZ1 and NVP-2, respectively, led to preferential and concentration-dependent downregulation of SE-related developmental and oncogenic transcription factor T (also known as brachyury/TBXT) in chordoma cells [80]. Furthermore, in acute monocytic leukemia cells, SY-1365, a recently developed highly selective inhibitor of CDK7 (currently under clinical trials in ovarian and breast cancer patients), preferentially reduced the expression of SE-related oncogenic transcription factors with a minor effect on housekeeping genes; these results support the idea that SE-driven genes are vulnerable to the inhibition of transcriptional CDKs [81]. Activating tumor suppressor p53 gene by casein kinase 1A1 (CKIα) inhibition is a potential approach for treating acute myeloid leukemia. Interestingly, A51, a small-molecule inhibitor co-targeting CKIα and CDK7/9 abolished several SE structures and preferentially suppressed the transcriptional elongation of SE-driven oncogenes, accompanied by synergistical stabilization of P53 [82]. Moreover, the favored loss of SE-associated transcription factors and DNA damage

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response genes was observed in T-cell acute lymphoblastic leukemia cells treated with THZ531, an inhibitor of CDK12 and CDK13, which also regulate transcriptional pol II elongation [83]. Although transcriptional CDK inhibitors are currently under evaluation in several clinical trials for multiple cancers [84], only a few studies have attempted to evaluate the effect of transcriptional CDK inhibitors in noncancerous inflammatory diseases [84]. Nevertheless, transcriptional CDKs have been implicated as playing a critical role in proinflammatory gene expression [85], suggesting the therapeutic potential of transcriptional CDK inhibitors in inflammatory diseases possibly through targeting SEs. Consistent with these reports, a recent study revealed that CDK7 inhibition by THZ1 ameliorated hyperinflammatory states in a preclinical model of cytokine release syndrome via preferential reduction of a set of inflammatory genes including IL1 and STAT1 that are associated with SEs [86].

Although clinical applications of BET and transcriptional CDK inhibitors in inflammatory diseases may be a little far from realization, there is no doubt regarding the potential therapeutic targets for enhancer/SE inhibition in various diseases, including inflammatory diseases.

Conclusion

Recent advances in sequencing- and microscopy-based technologies have unveiled the hidden functions of enhancers residing across most of the noncoding human genome. SEs normally control the expression of transcripts that play prominent roles in cell-type-specific processes. Beyond fundamental roles of SEs in the maintenance of cell identity and the determination of cell fate, the significance of SEs has been implicated in external perturbations including inflammation. Notably, although SEs apparently act as multimolecular assemblies, SEs can be newly formed within a short period in response to inflammatory stimuli, resulting in a rapid transition from a resting to an inflammatory transcription program. Moreover, the development of small-molecule inhibitors targeting enhancers has contributed to a better understanding of enhancer/SE functions and highlighted their association with diseases. However, current understanding of enhancer biology is still constrained by various uncertainties and rather limited to the research fields of cancer and development/differentiation. For example, recent studies have suggested that SEs participate in forming phase-separated condensates for maximizing transcriptional output; however, it remains unknown whether this type of model could be utilized in signal-dependent transcription mechanisms and whether small molecules that disrupt phase-separated condensates could serve as potential candidates for targeting enhancers/SEs and their associated diseases (Fig. 3). Further studies are warranted for understanding the mechanisms through which enhancer/SE precisely mediate transcriptional activation during response to external perturbations, including inflammation.

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Conflict of interest

The authors declare no conflict of interest.

Author contributions

YH and YK conceptualized the review, searched the relevant literature, wrote the manuscript, and created the figures.

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